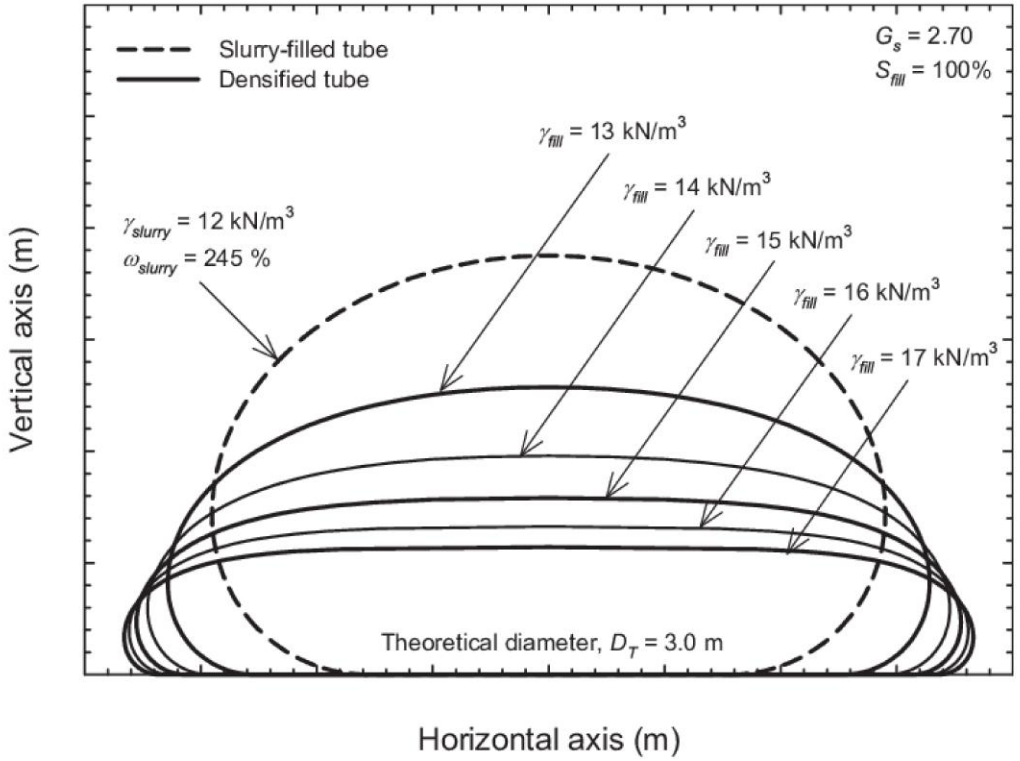
Smart Coast - Wave - Weave

Fiber optics for breakwater deformation sensing application

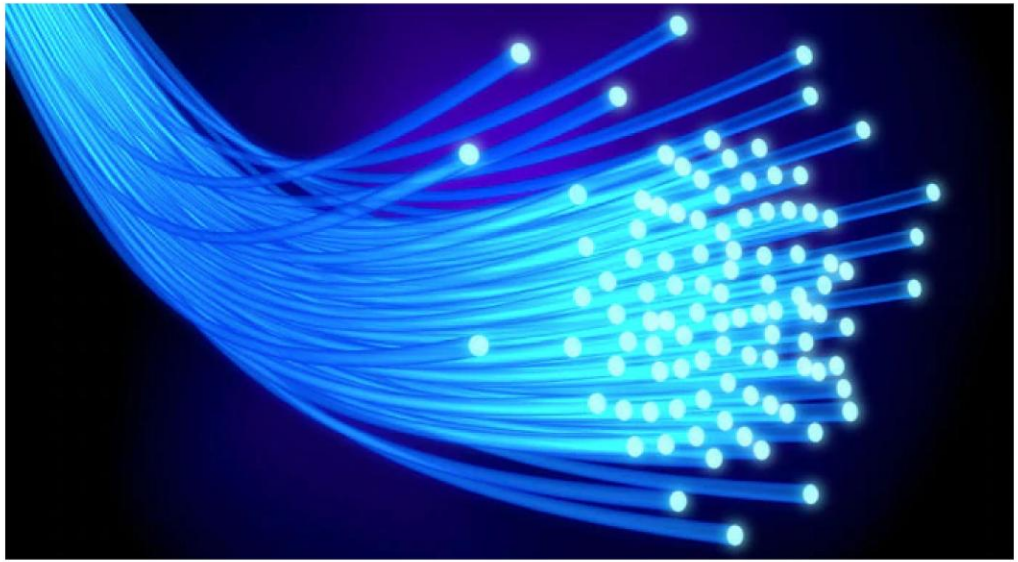




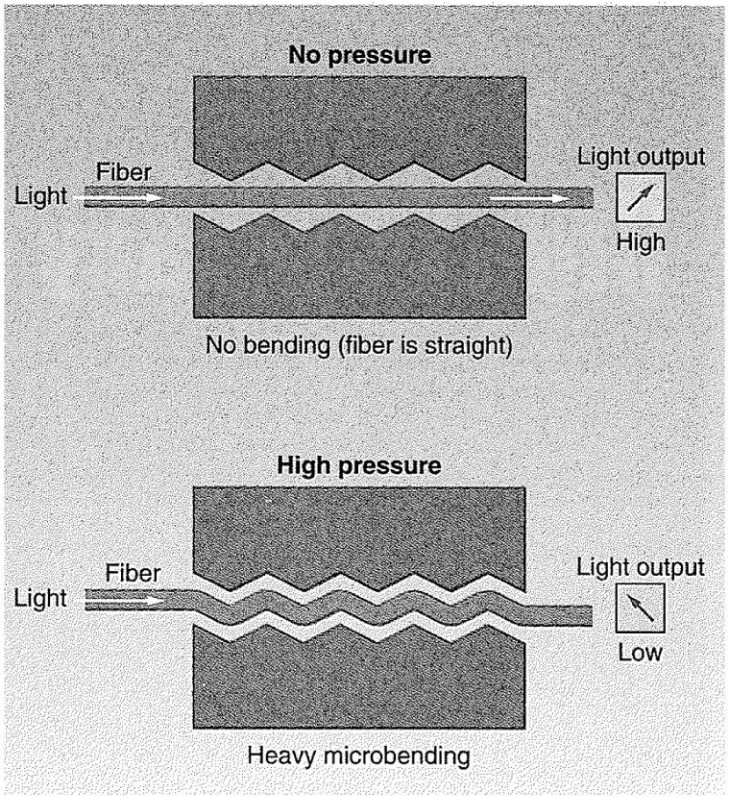


Geotextile fabric

1 Choice of Fiber Optics

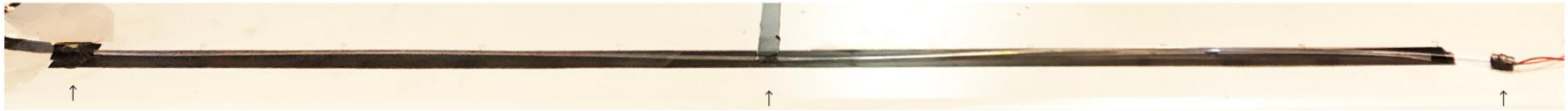


2 Working Principles



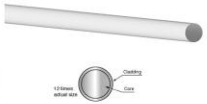
Hecht, J. 1999. "Fiber Sensors Capture Environmental Changes."

3.1 Fiber Optic Setup





RGB LED

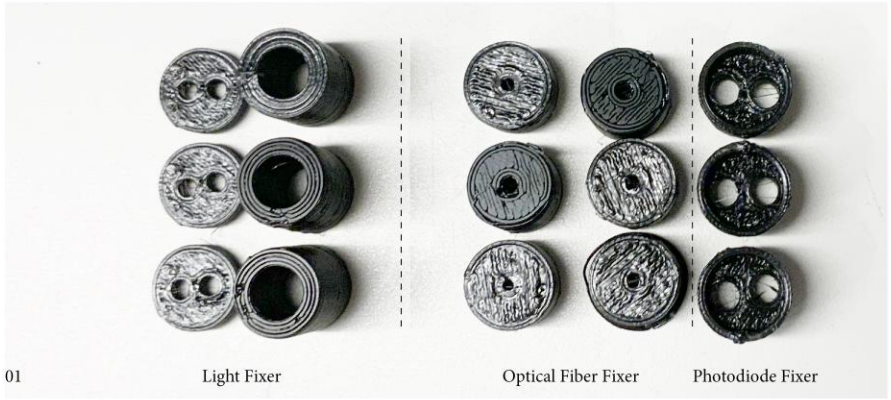


POF, Step Index, 3000 µm Core
MODEL: F-POF-3000

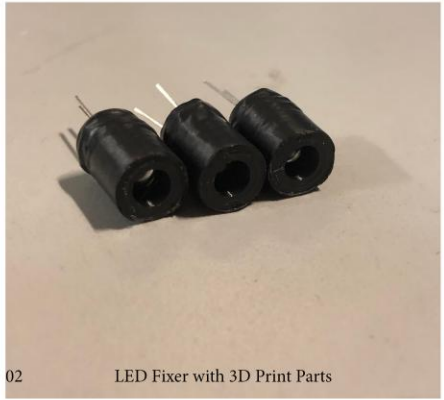


BPW 34

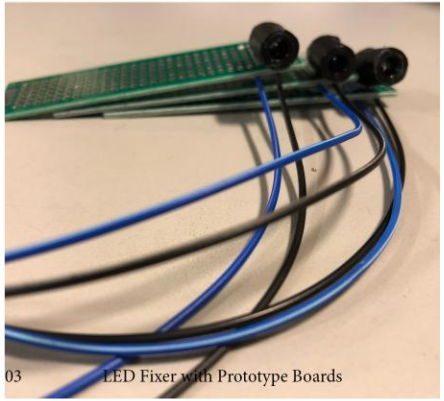
3.1 Fiber Optic Setup - Connection



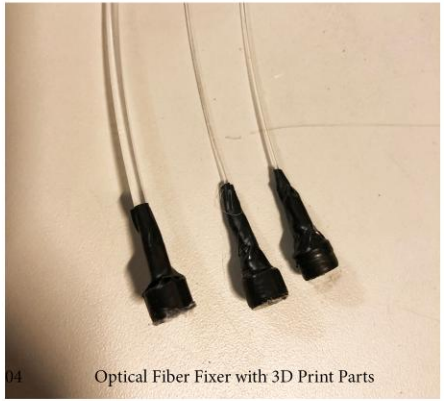
01 Light Fixer Optical Fiber Fixer Photodiode Fixer



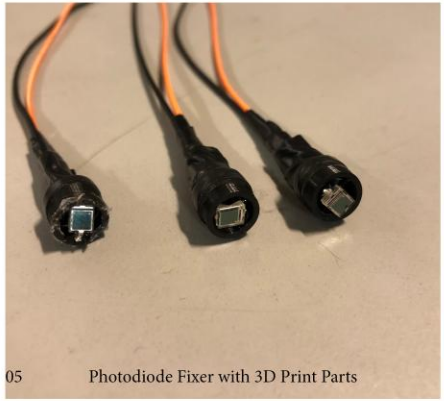
02 LED Fixer with 3D Print Parts



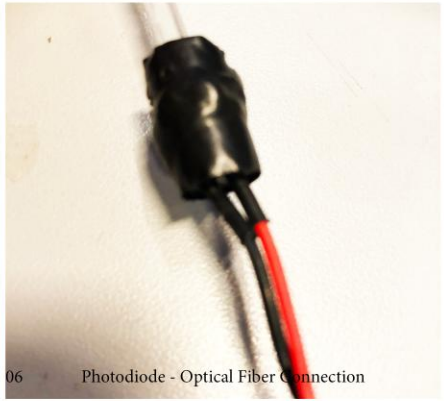
03 LED Fixer with Prototype Boards



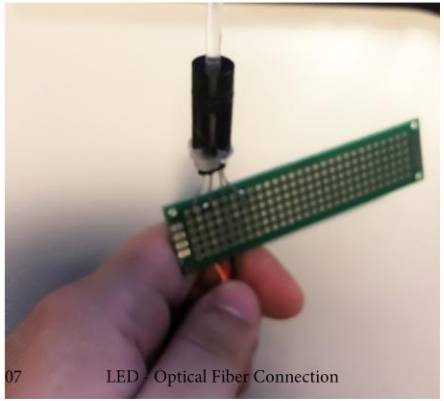
04 Optical Fiber Fixer with 3D Print Parts



05 Photodiode Fixer with 3D Print Parts

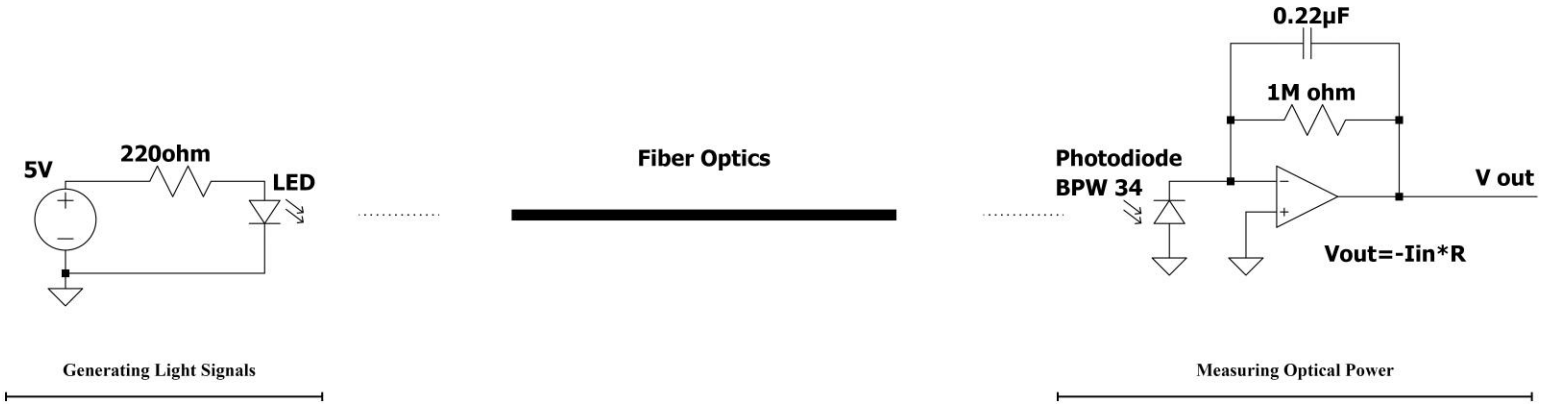


06 Photodiode - Optical Fiber Connection



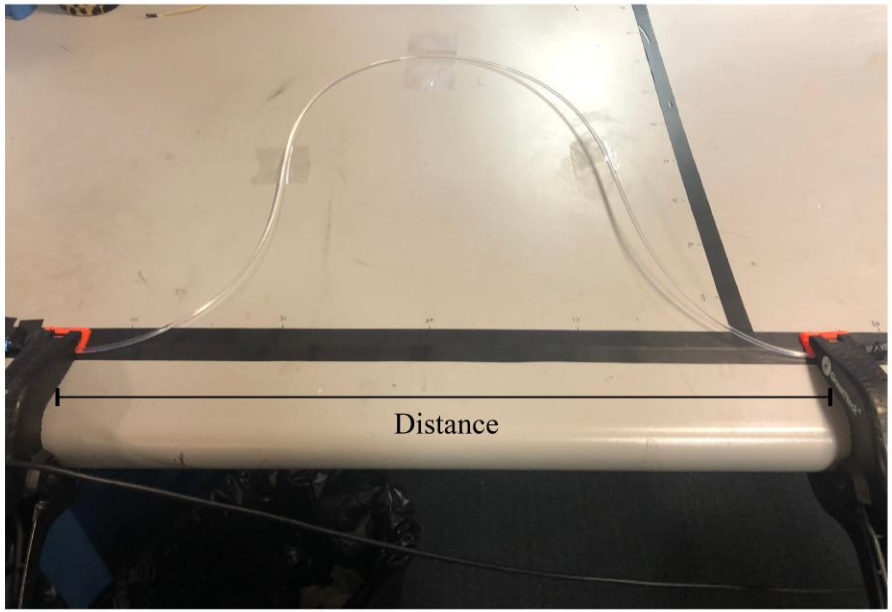
07 LED - Optical Fiber Connection

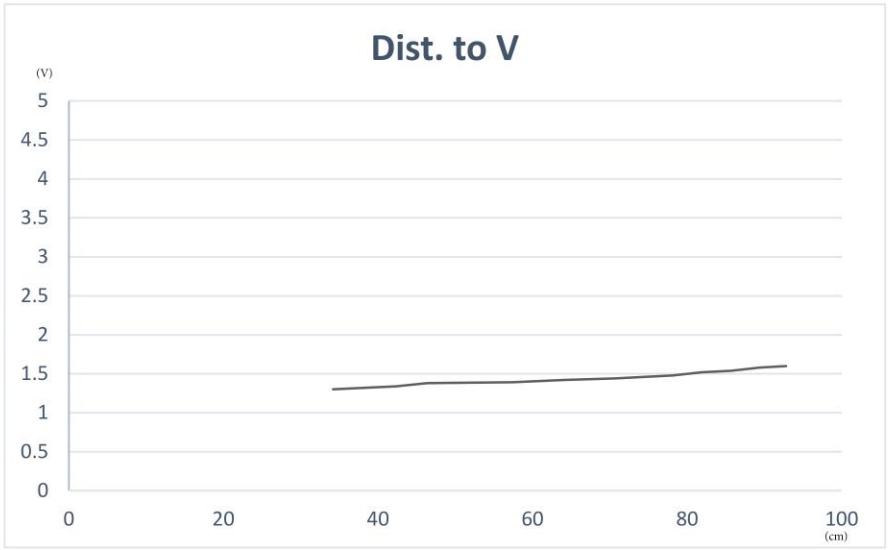
3.2 Circuit


5V
220ohm
LED
Fiber Optics
0.22µF
1M ohm
Photodiode
BPW 34
V out
Vout=-Iin*R
Generating Light Signals
Measuring Optical Power

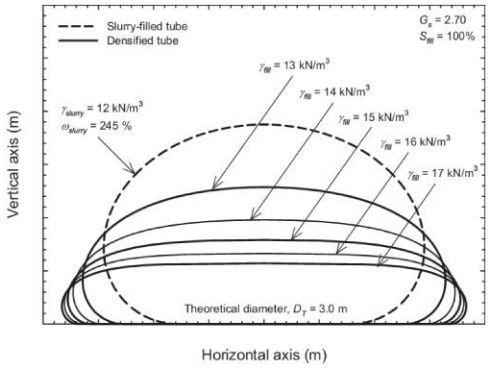
4.1 Bending Experiment

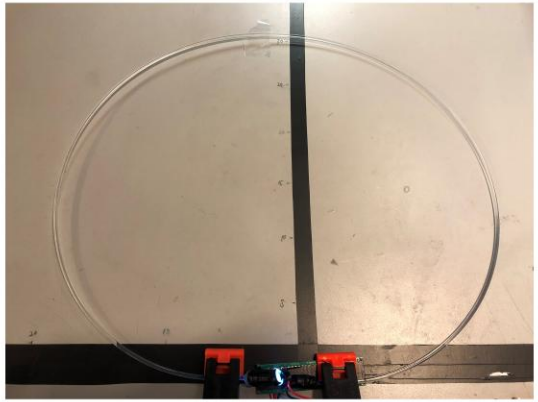
Set-up

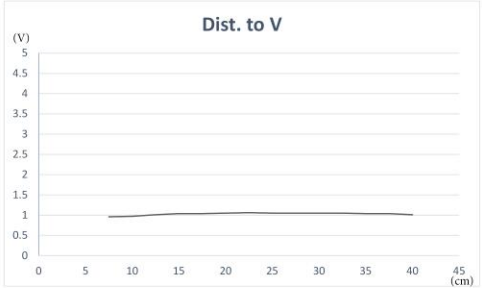


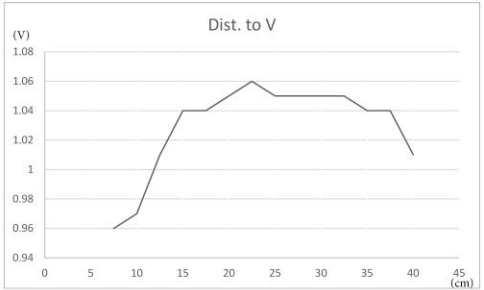


4.1 Bending Experiment 2

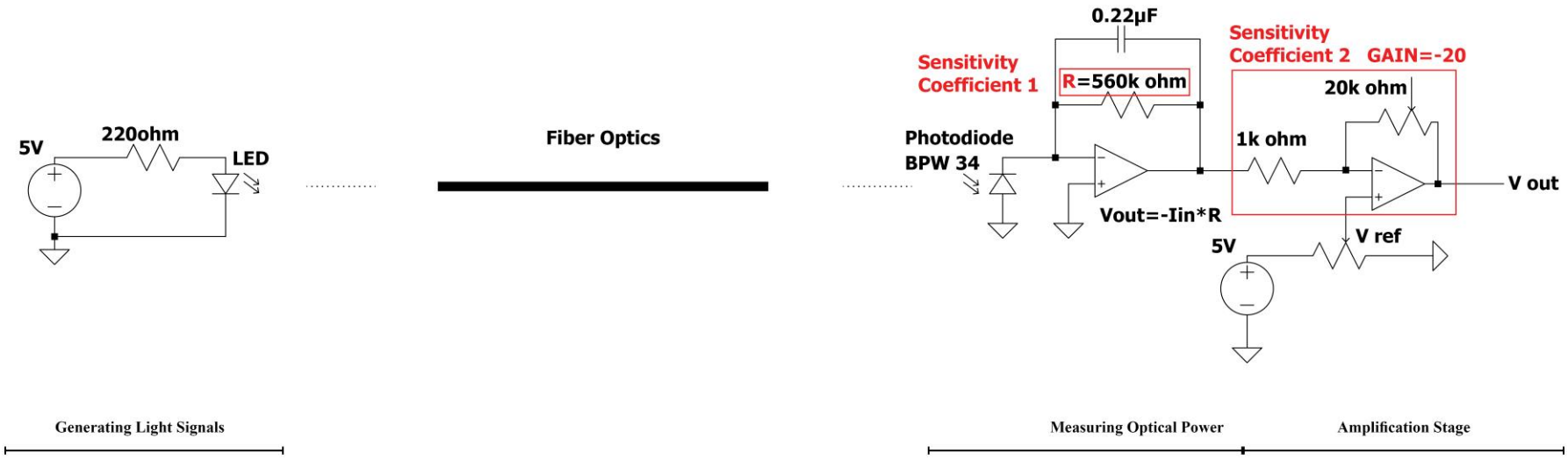


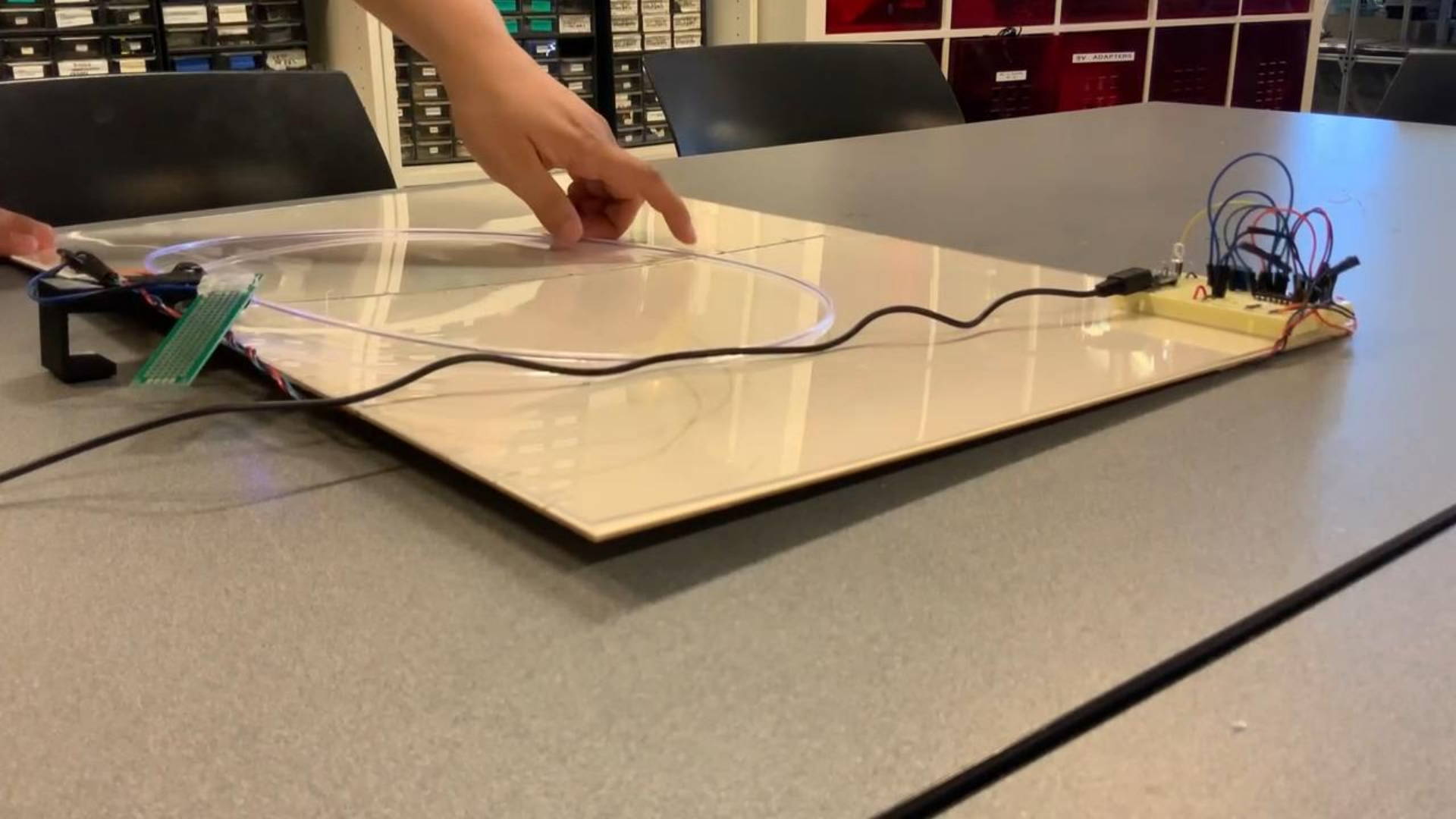




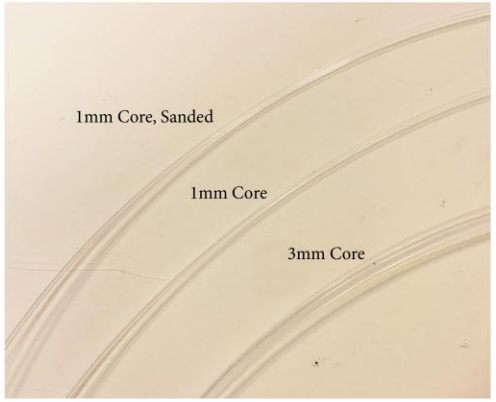


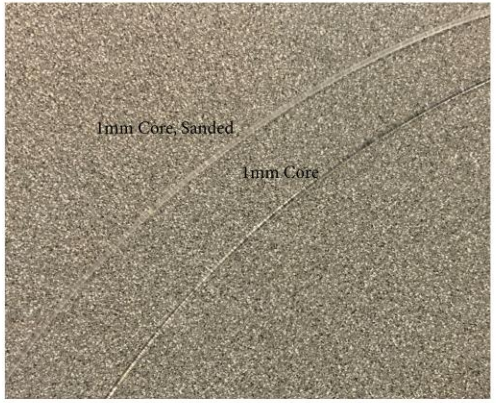
4.2 Improving Sensitivity - Amplified Circuit

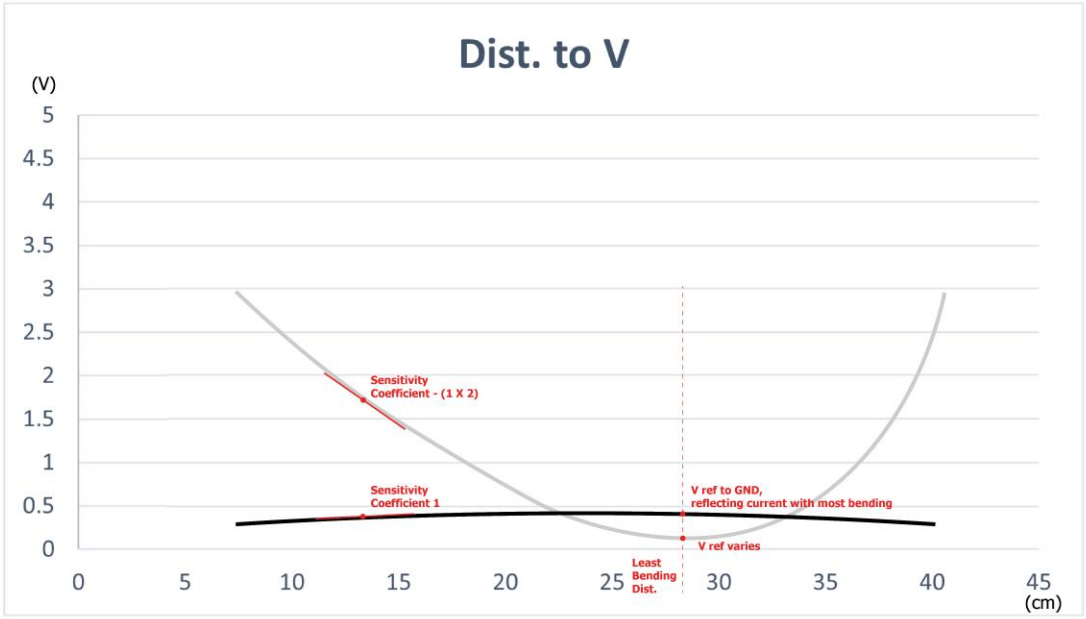




4.2 Improving Sensitivity - Graph Explanation





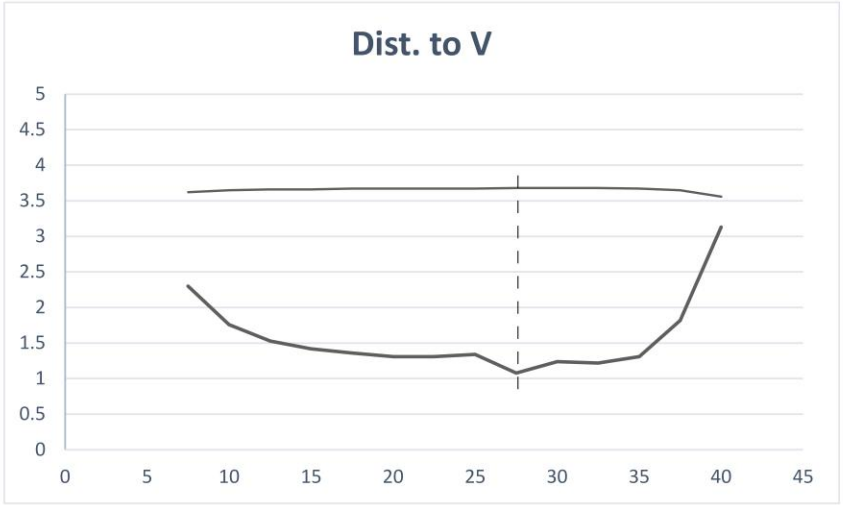


- **V- Shape Curve**

4.2 Improving Sensitivity - Fiber Optic - Fiber Thickness

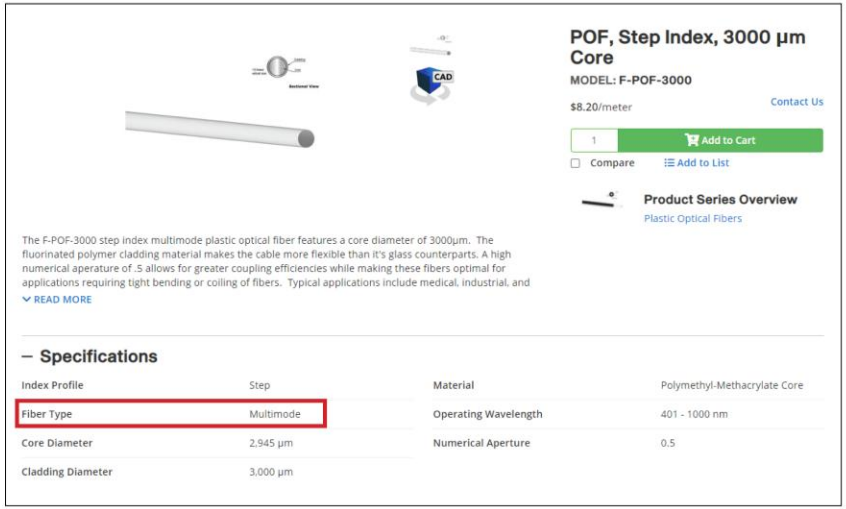


↑ **3mm Fiber**
Blue LED
R= 560k ohm saturated, 267k ohm



↓ **1mm Fiber**
Blue LED
R=560k ohm

4.2 Improving Sensitivity - Fiber Optic - Fiber Thickness



POF, Step Index, 3000 µm Core

MODEL: F-POF-3000

$8.20/meter Contact Us

1 Add to Cart

Compare Add to List

Product Series Overview

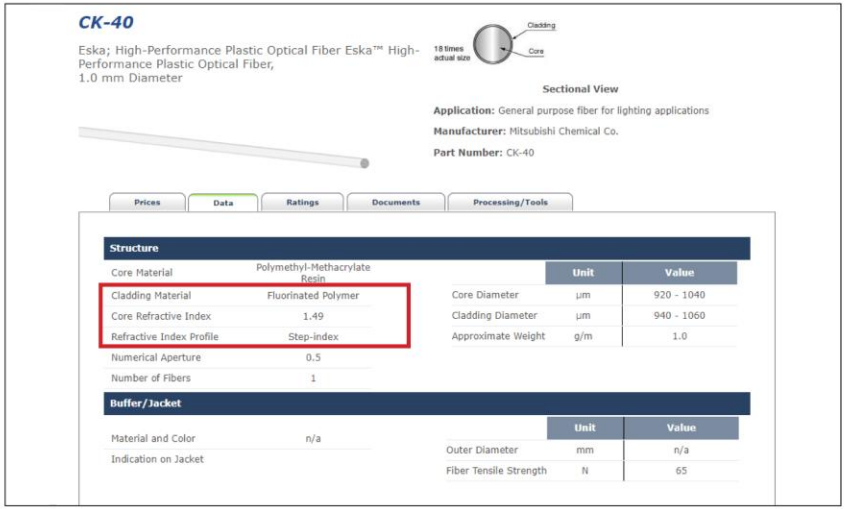
Plastic Optical Fibers

The F-POF-3000 step index multimode plastic optical fiber features a core diameter of 3000µm. The fluorinated polymer cladding material makes the cable more flexible than it's glass counterparts. A high numerical aperture of .5 allows for greater coupling efficiencies while making these fibers optimal for applications requiring tight bending or coiling of fibers. Typical applications include medical, industrial, and

READ MORE

– Specifications

Index Profile	Step	Material	Polymethyl-Methacrylate Core
Fiber Type	Multimode	Operating Wavelength	401 - 1000 nm
Core Diameter	2,945 µm	Numerical Aperture	0.5
Cladding Diameter	3,000 µm		



CK-40

Eska; High-Performance Plastic Optical Fiber Eska™ High-Performance Plastic Optical Fiber, 1.0 mm Diameter

Cladding
Core
18 times actual size

Sectional View

Application: General purpose fiber for lighting applications

Manufacturer: Mitsubishi Chemical Co.

Part Number: CK-40

Prices | Data | Ratings | Documents | Processing/Tools

Structure

Core Material	Polymethyl-Methacrylate Resin
Cladding Material	Fluorinated Polymer
Core Refractive Index	1.49
Refractive Index Profile	Step-index
Numerical Aperture	0.5
Number of Fibers	1

	Unit	Value
Core Diameter	µm	920 - 1040
Cladding Diameter	µm	940 - 1060
Approximate Weight	g/m	1.0

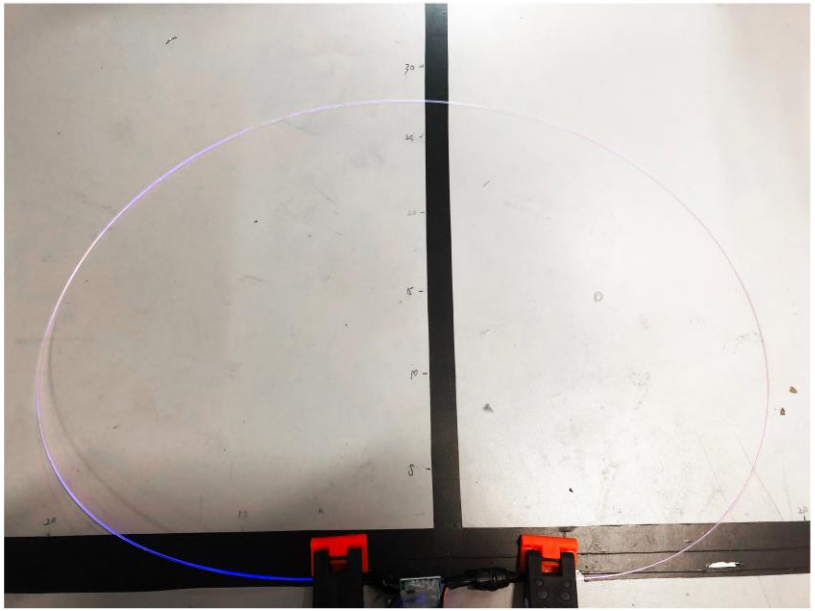
Buffer/Jacket

Material and Color	n/a
Indication on Jacket	

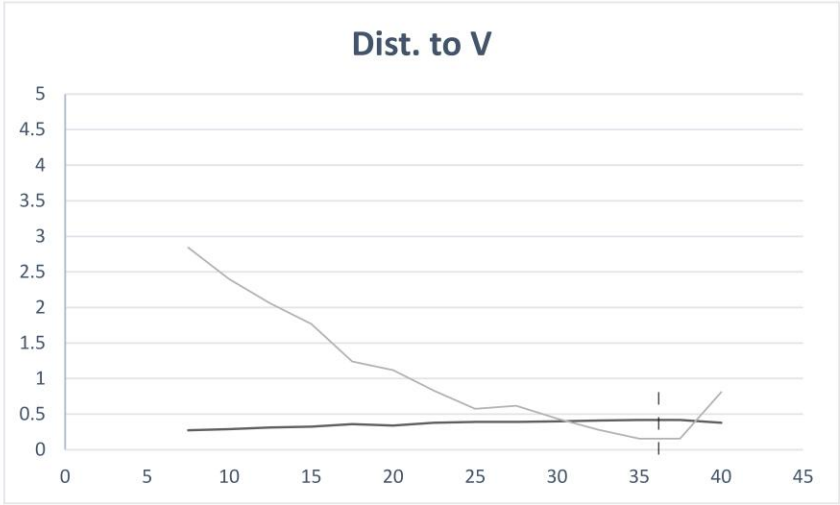
	Unit	Value
Outer Diameter	mm	n/a
Fiber Tensile Strength	N	65

- **Contact area** of the photodiode, LED to optical fibers is likely to affect most sensitivity

4.2 Improving Sensitivity - Fiber Optic - Sanded Fiber



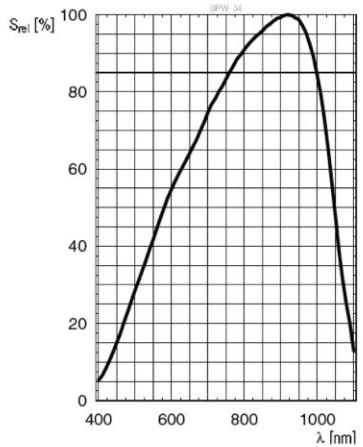
- Side glow!
- Light luminance received reduced
- Sensitivity improved
- Drifting max. bending dist. to be varified



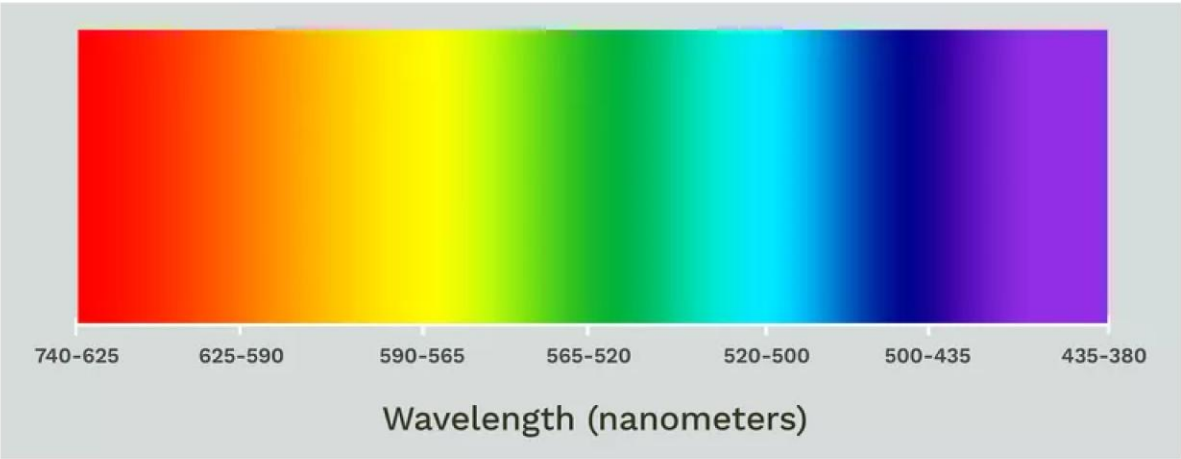
↑ **1mm Fiber, Sanded**
Blue LED
R= 560kohm
V bias=1.0V

4.2 Improving Sensitivity - Photodiode - Wavelength of the Light

Photodiode Datasheet
Relative Spectral Sensitivity



Visible Light Spectrum and RGB



- Photodiode spectrum sensitivity, optical fiber operating wavelength, environmental circumstances

4.2 Improving Sensitivity

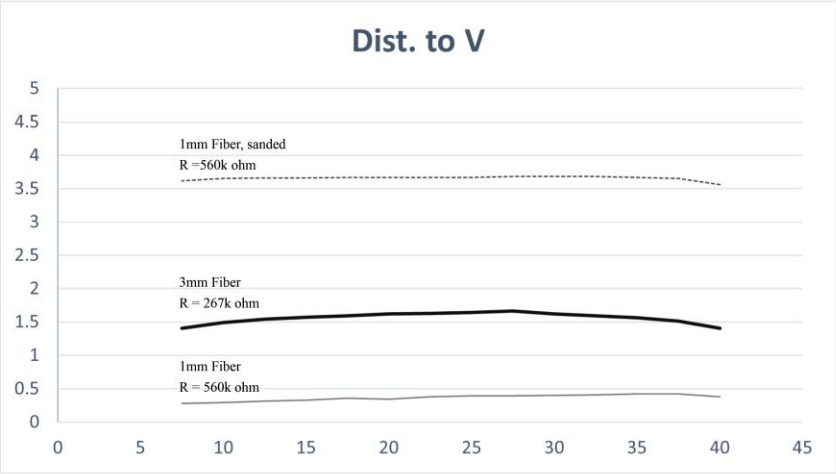


↑ 3mm Fiber
White LED
R= 560k ohm

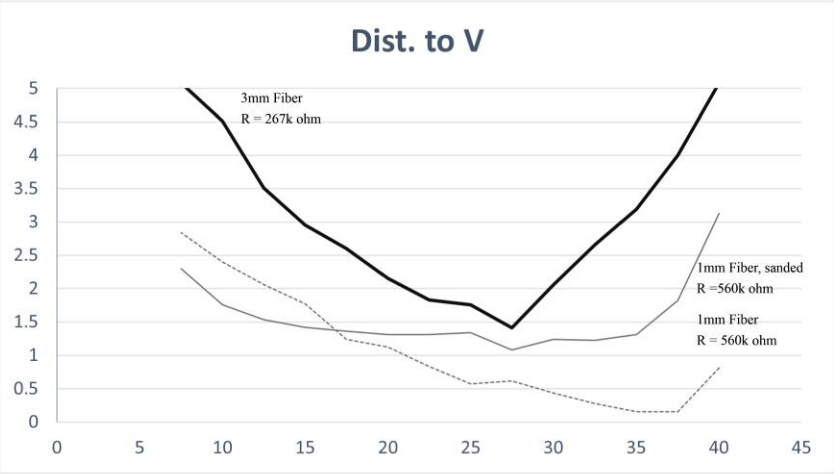


↓ 3mm Fiber
Blue LED
R= 560k ohm saturated,
267k ohm

4.2 Sensitivity Comparison



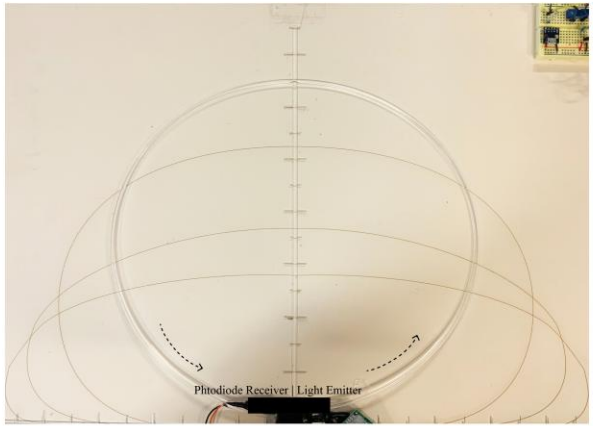
First Stage, I-to-V



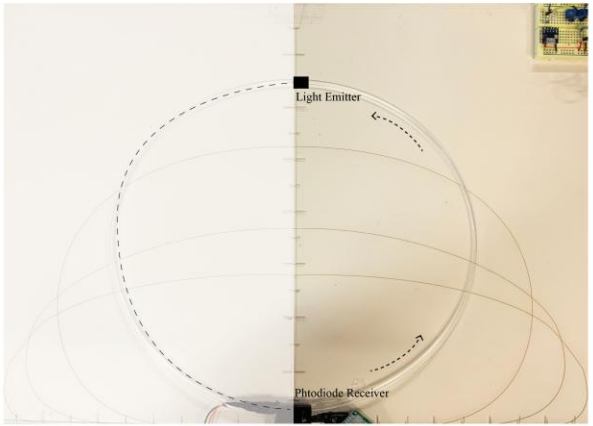
Second Stage, Inverting Amplification
(GAIN = -20)

• Sensitivity: 3mm Fiber >> 1mm Fiber, sanded >> 1mm Fiber

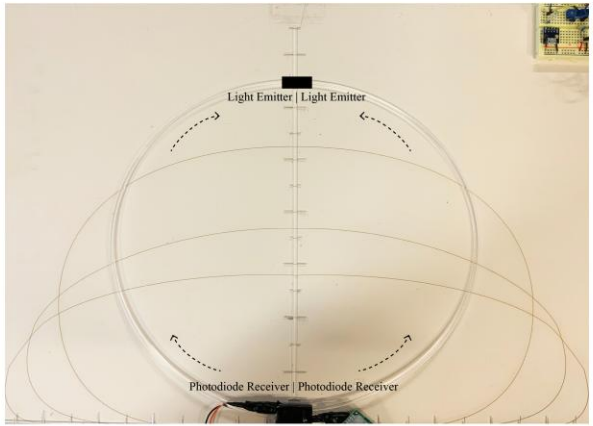
4.3 Weaving of the Geotube



Round Weaving

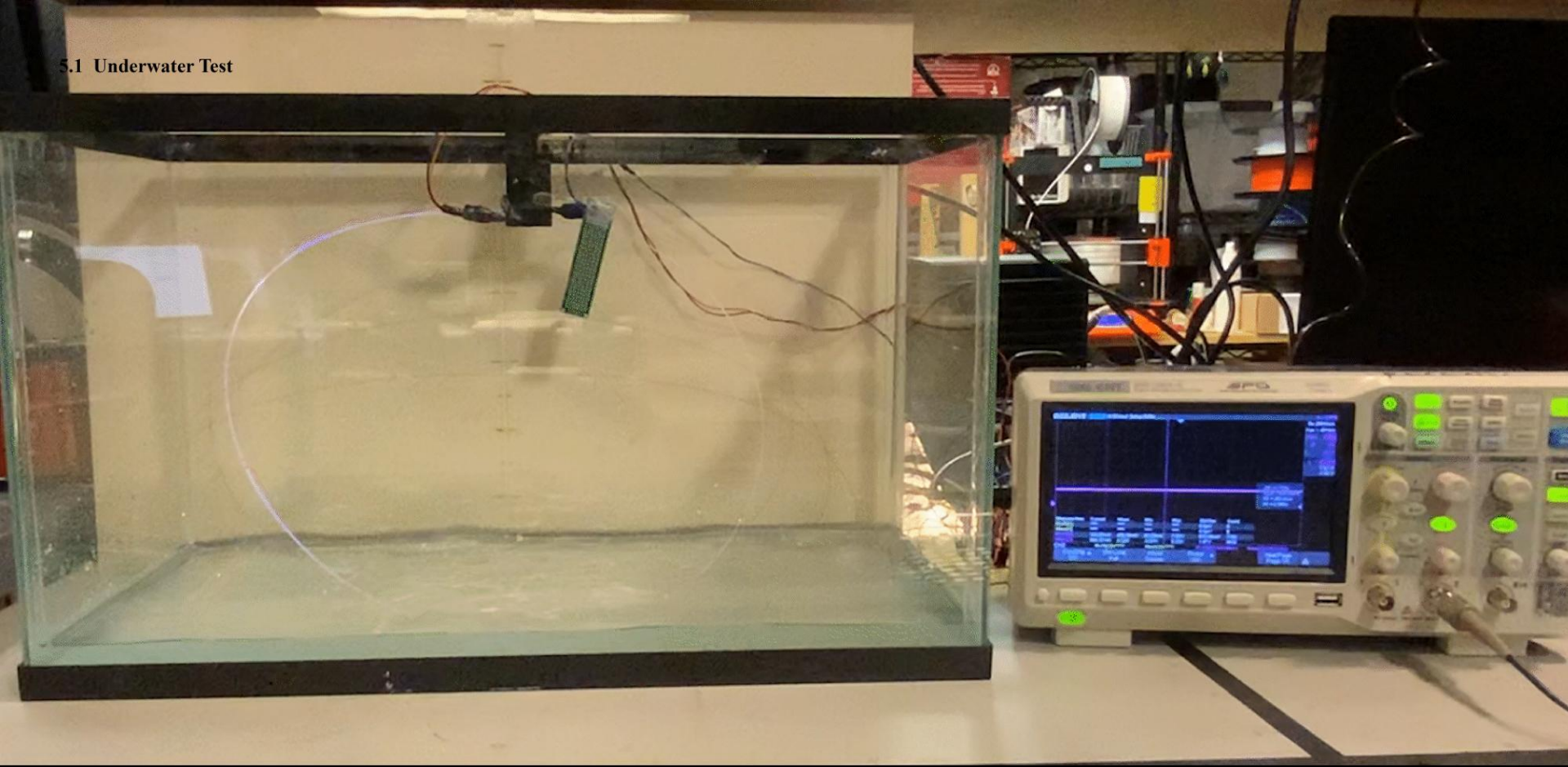


Half Weaving on One Side

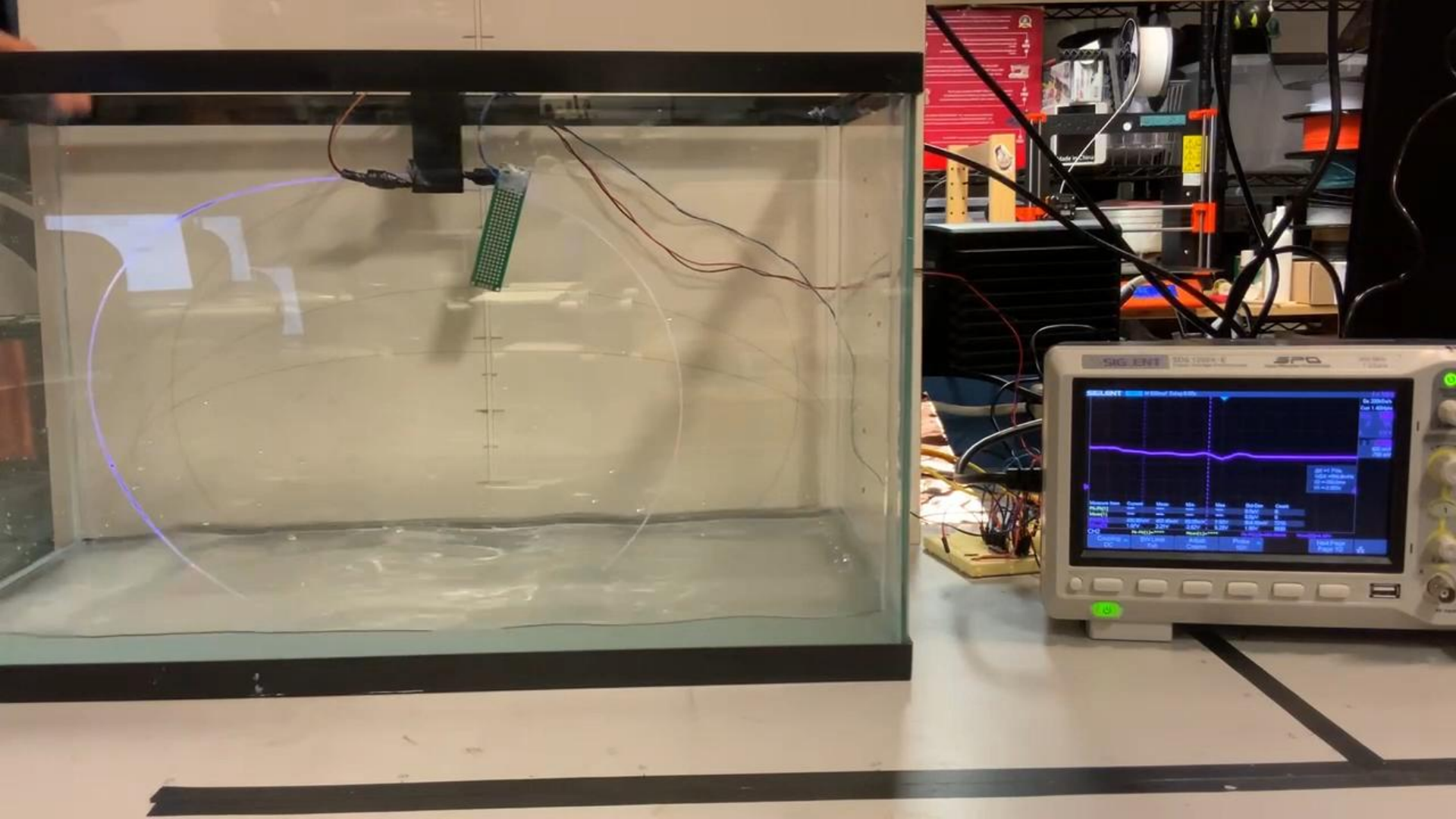


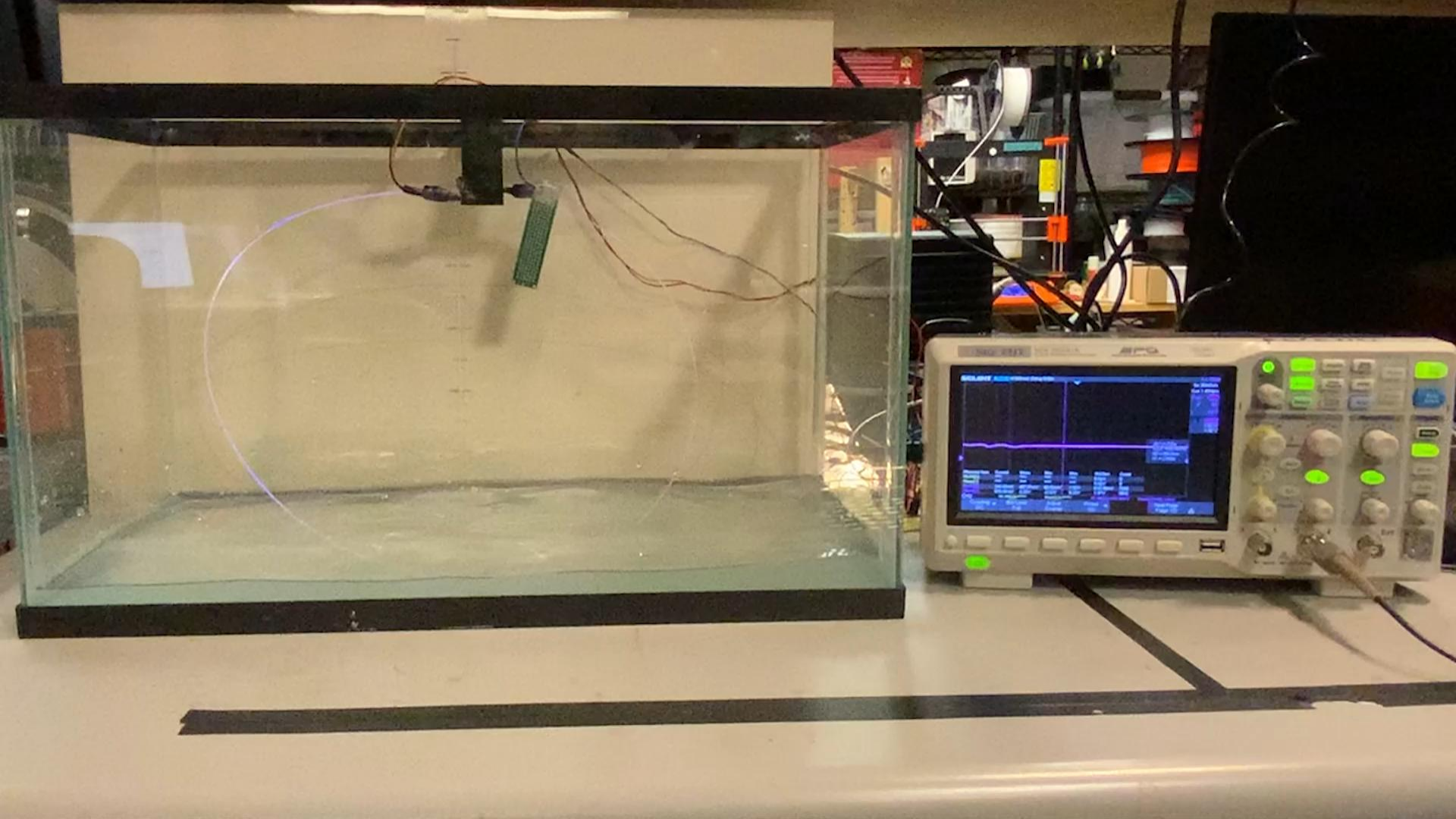
Half Weaving on Both Side

5.1 Underwater Test

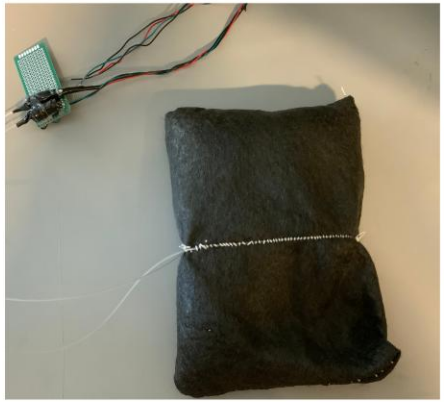


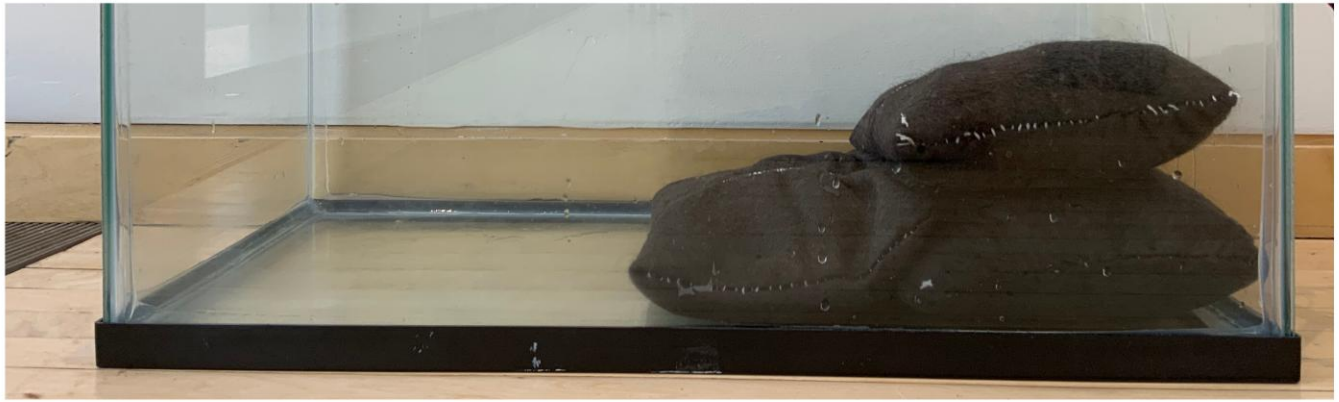
Refraction rates difference of the medium, water and air. did not generate a noticable change in signals.

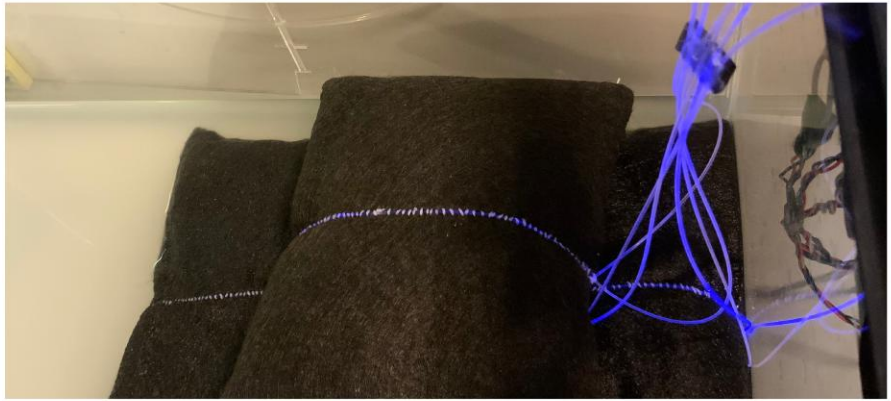


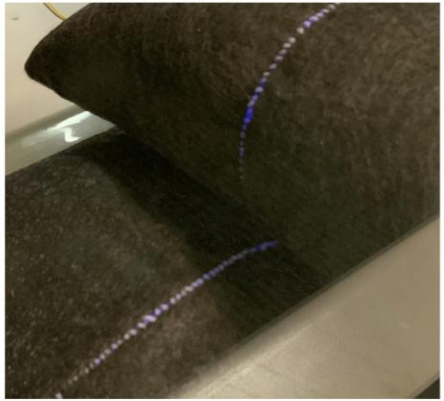


5.2 Breakwater Weaving Experiment



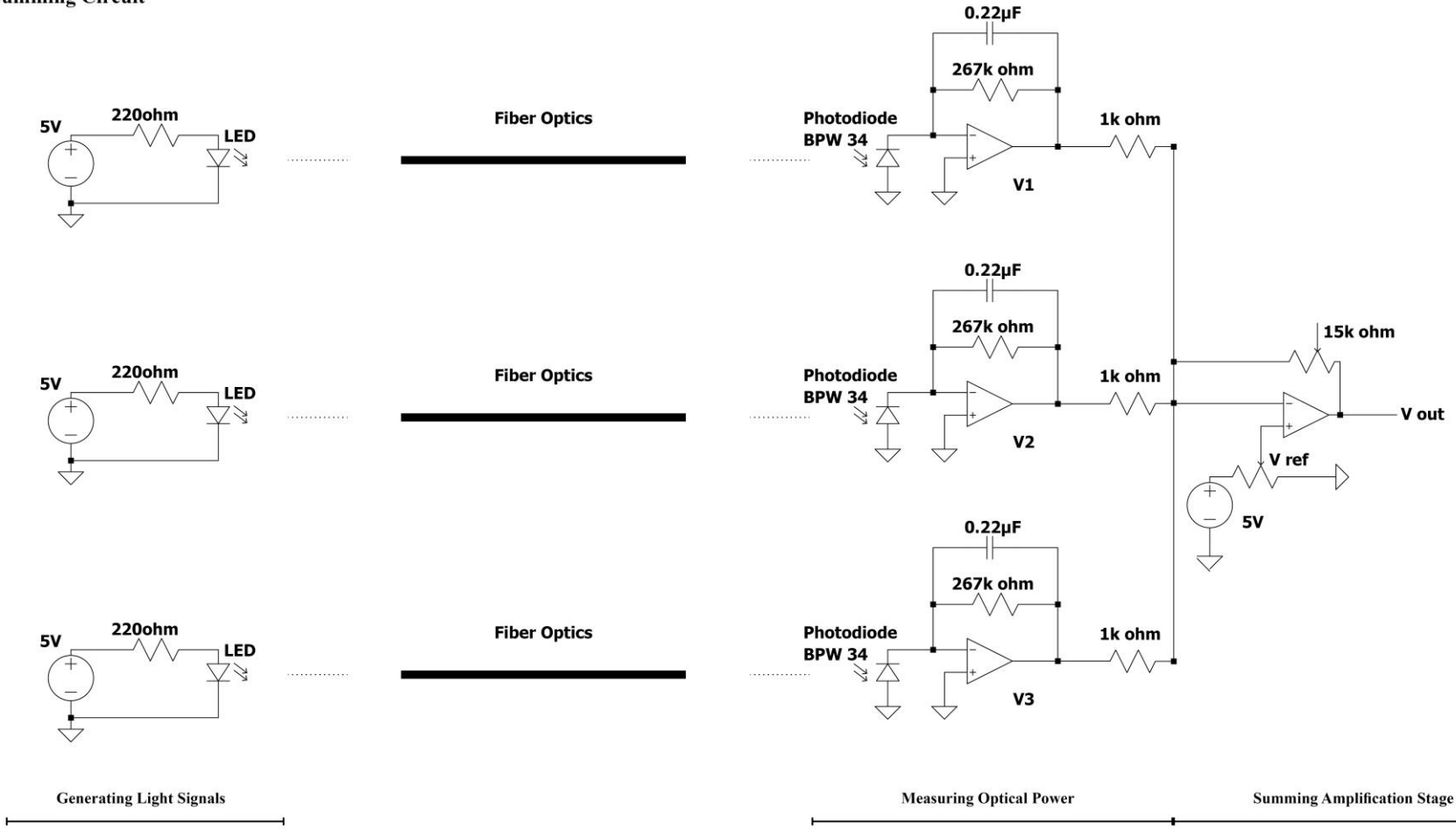








5.2 Summing Circuit



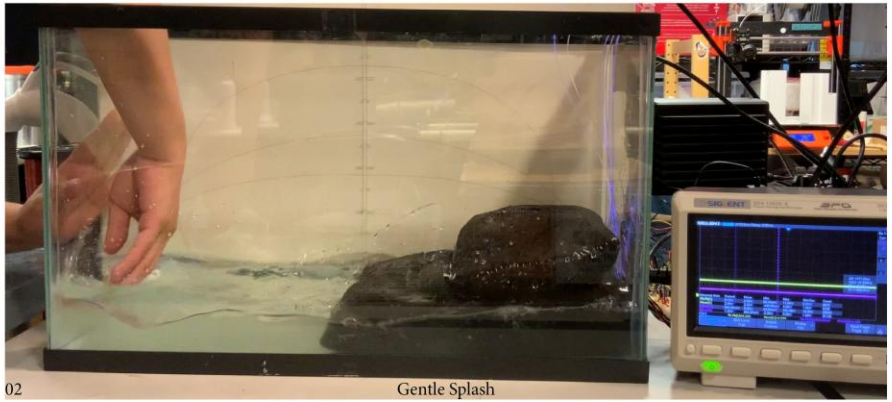

CLASS WORKBENCH - KEEP CLEAR

5.3 Data Behavior

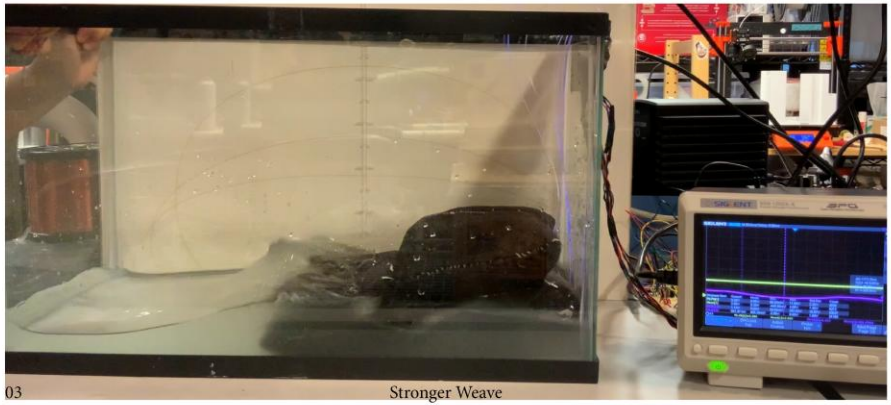
• For long term simulation, will the data reflects the amplitude of the wave?



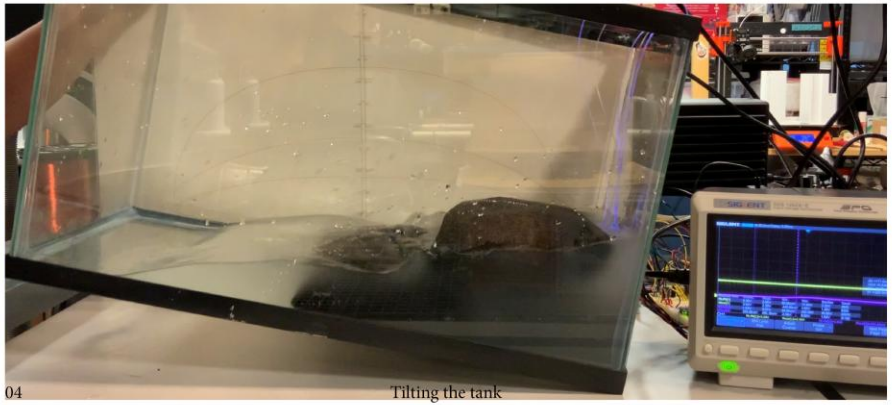
01 Pouring Water



02 Gentle Splash



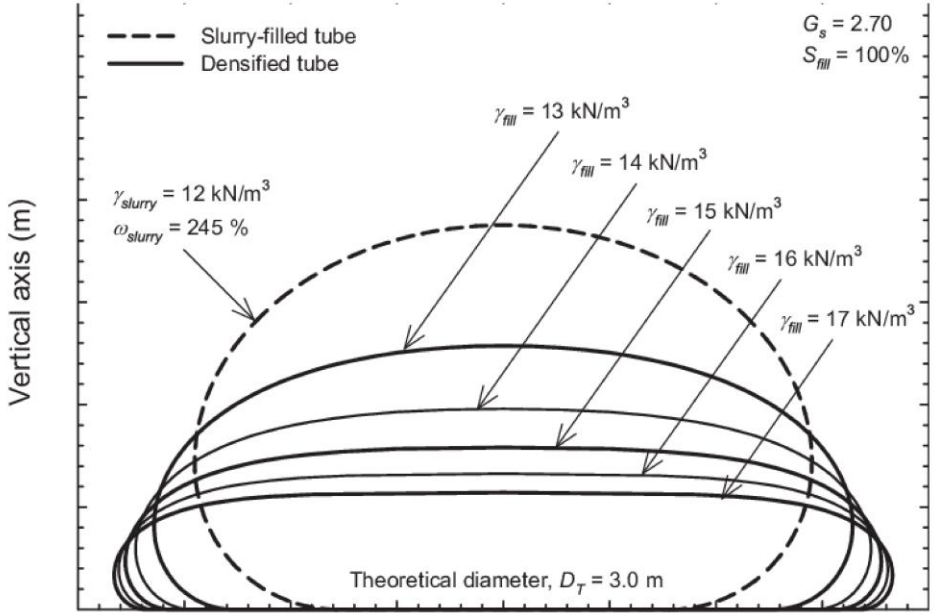
03 Stronger Weave



04 Tilting the tank

6 Expectation on the Deformation Behavior of the Breakwater Bag

Cause of Deformation		Data Period		
• Material Property inside the Bag	→	• Weekly, Monthly	→	• Low Pass Filter
• Wave Pressure	→	• Transient Data	→	• High Pass Filter



7 Process Reflection and Next Step

Rework on

- Connection, connection, connection
- Rigorous bending fiber optic sensor test
- Exploration of the best wavelength of the light

Next Step

- Wave simulation with breakwater bags and decoding data

• Sincere thanks to the whole Responsive Environments Group

Thanks to Joe for this dense and amazing class, and let it happen again before I graduate;
Thanks to Fangzheng for the most timely and constant help, organization, and presence at every recitation and class;
Thanks to Mark for all the electronics help;
Thanks to Juliana for introducing fiber optics to me and being so inspiring as women model;
Thanks to Patrick for introducing the group to my last semester and answering me every environmental and citizen sensing conceptual question;
Thanks to Brian for making the Tidmarsh happen, which led me into the fascination with environmental sensing;
Thanks to Cedric for all the setup suggestions;
Thanks to David, Irmandy, Devora, Donald, Ali, Sam, for all the amazing works and guest talks in class and recitations. I wish I could spend more time here.

